

Lecture 10

More on linear regression

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

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OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Homework 2 is due Monday before class

Plan for today

- Why can $X^T X$ be (practically) singular? 8 min
- What to do about it: Pseudoinverse 8 min
- What to do about it: Gradient descent 10 min
- Why squared error? 15 min
- Ridge regression 10 min

Recall: Linear regression problem

Problem (Linear least squares regression)

Given examples $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^p \times \mathbb{R}$, find $\theta \in \mathbb{R}^p$ minimizing

$$\text{MSE}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta^\top x_i)^2.$$

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Idea: Minimizers $\hat{\theta}$ solve normal equation $X^\top X \hat{\theta} = X^\top y$. When $(X^\top X)^{-1}$ exists,

$$\hat{\theta} = (X^\top X)^{-1} X^\top y \quad \text{hence} \quad \hat{y} = X \hat{\theta} = X(X^\top X)^{-1} X^\top y.$$

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Question (s)

What if $X^\top X$ is (practically) singular? Why *squared* error?

Why $X^T X$ can be singular

Proposition

If one feature (a column of X) is a linear combination of the others, then $X^T X$ is singular.

Proof. Say $X_p = c_1 X_1 + \dots + c_{p-1} X_{p-1}$. Then $v = (c_1, \dots, c_{p-1}, -1) \neq 0$ has $Xv = 0$, so $X^T X v = 0$. A matrix that sends a nonzero vector to 0 has no inverse. □

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Watch out: Even if X_p is only **nearly** such a combination, $X^T X$ will be nearly singular, making the calculation of $(X^T X)^{-1}$ prone to errors.

Example: A repeated direction

Suppose the design matrix has the form

$$X = \begin{bmatrix} 1 & x & 2x \end{bmatrix},$$

so $p = 3$ but $\text{rank}(X) = 2$. Then

$$X\theta = \theta_1 \mathbf{1} + (\theta_2 + 2\theta_3) x.$$

So the “fit” only sees $\theta_2 + 2\theta_3$. There are infinitely many $\hat{\theta}$ that minimize the MSE and produce the same $\hat{y}$.

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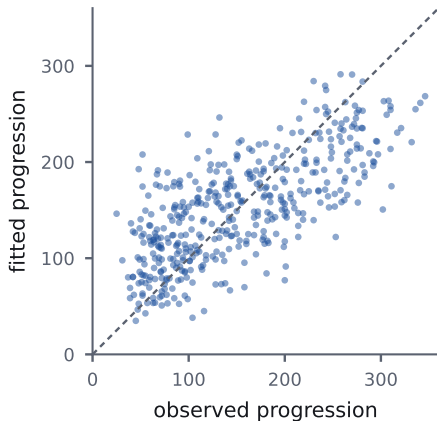
So the “fit” only sees $\theta_2 + 2\theta_3$. There are infinitely many $\hat{\theta}$ that minimize the MSE and produce the same $\hat{y}$.

Watch out: The fit is unaffected, but the coefficients are not determined at all. We could set $\theta_2 = 0$ or $\theta_3 = 0$, making the second or third predictors seem useless.

Recall: Disease progression data

From $n = 442$ diabetes patients:

- each x_i has 10 measurements:
 1. age
 2. blood pressure
 3. body mass index
 4. sex
 - 5–10. six blood serum measurements (“s1” through “s6”)
- the response y_i : a measure of disease progression one year later



Efron, Hastie, Johnstone and Tibshirani (2004)

Example: Quadratic features

Efron et al. fit their disease progression data with 64 features:

- the 10 measurements, standardized
- all 45 products of two of them, e.g. `bmi` · `bp`, `s1` · `s5`
- the 9 squares aside from that of `sex`, e.g. `bmi^2`

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$$s2 = TC - HDL - TG/5 = s1 - s3 - \exp(s5).$$

Key idea: Since $\exp(s5) \approx 1 + s5 + \frac{1}{2}s5^2$, $X^T X$ is nearly singular!

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Definition: Pseudoinverse

Definition (Moore–Penrose pseudoinverse)

The pseudoinverse of a matrix $X \in \mathbb{R}^{n \times p}$ with rank r and (compact) singular value decomposition $X = U_r \Sigma_r V_r^T$ is

$$X^+ = V_r \Sigma_r^{-1} U_r^T = \sum_{j=1}^r \frac{1}{\sigma_j} v_j u_j^T \in \mathbb{R}^{p \times n}.$$

- The columns of $U_r \in \mathbb{R}^{n \times r}$ form an orthonormal basis for the column space of X , so $U_r U_r^T$ projects onto it!

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- When X has full column rank, $XX^+ = X(X^T X)^{-1} X^T$ anyway (!!!)

Theorem: Pseudoinverse solution of least squares

Theorem (Least squares via the pseudoinverse)

Let $X \in \mathbb{R}^{n \times p}$ be a rank- r design matrix with SVD $\sum_{j=1}^r \sigma_j u_j v_j^\top$ and let $y \in \mathbb{R}^n$ be the response. Then,

$$\theta^+ = X^+ y = \sum_{j=1}^r \frac{1}{\sigma_j} v_j (u_j^\top y)$$

1. minimizes $\text{MSE}(\theta)$
2. is the minimizer of smallest norm, and
3. is the only minimizer when $r = p$.

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Complexity of solving the normal equation

To solve $X^T X \hat{\theta} = X^T y$ we must form $X^T X$ and then factor it:

$$\underbrace{np^2}_{\text{form } X^T X} + \underbrace{p^3}_{\text{factor it}} \quad \text{arithmetic operations.}$$

- at $n = 10^6$ and $p = 10^4$, forming $X^T X$ alone is 10^{14} operations
- and $X^T X$ is $10^4 \times 10^4$, which has to be stored

Idea: Computing $\nabla \text{MSE}(\theta) = \frac{2}{n} X^T (X\theta - y)$ is two matrix-vector products, or $2np$ operations, which is 2×10^{10} here. We could calculate the gradient 5000 times for the price of forming $X^T X$ once.

Algorithm: Gradient descent

Definition (Gradient descent for least squares)

Choose $\theta^{(0)}$ and a step size $\eta > 0$, and repeat

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \text{MSE}(\theta^{(t)}) = \theta^{(t)} - \frac{2\eta}{n} X^\top (X\theta^{(t)} - y).$$

- each step uses one product $X\theta^{(t)}$ and one product with $X^\top$

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- each step uses one product $X\theta^{(t)}$ and one product with X^T
- we never need to form $X^T X$
- we get to choose η , but not all choices work

Why **gradient descent** minimizes MSE

- If $\hat{\theta}$ solves the normal equations then $\nabla \text{MSE}(\theta) = \frac{2}{n} X^\top X (\theta - \hat{\theta})$, so subtracting $\hat{\theta}$ from both sides of the update gives

$$\theta^{(t+1)} - \hat{\theta} = \left(I - \frac{2\eta}{n} X^\top X \right) (\theta^{(t)} - \hat{\theta}).$$

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- In an orthonormal eigenbasis of $X^T X$, the component with eigenvalue λ is multiplied by $1 - 2\eta\lambda/n$ each step. It shrinks exactly when

$$\left| 1 - \frac{2\eta\lambda}{n} \right| < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{n}{\lambda},$$

and this must hold for every λ , so $\lambda_{\max}$ decides: $0 < \eta < n/\lambda_{\max}$.

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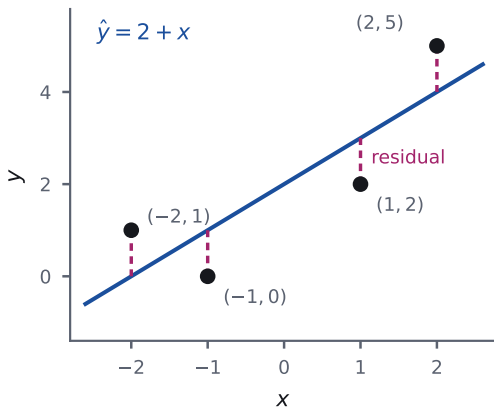
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- MSE is convex, so the limit is a **global** minimum

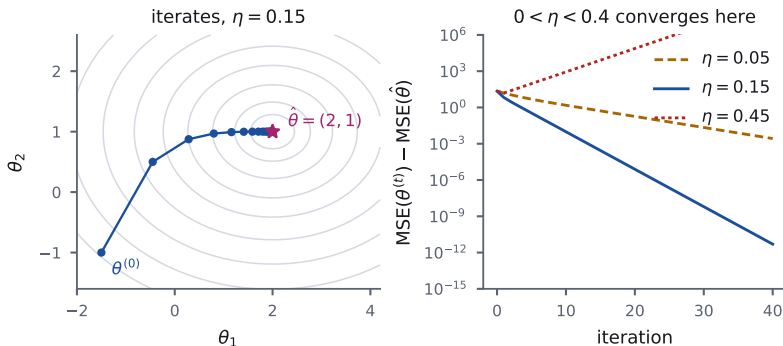
Example: Four points again



Example of four points from lecture 9.

Example: Four points again

$X^T X = \text{diag}(4, 10)$ and $n = 4$, so the iterates converge when $0 < \eta < 0.4$. At $\eta = 0.15$ the factors are 0.70 and 0.25, and $\text{MSE} = 1.0098$ after ten steps against the exact 1.



What gradient descent is really for

Here we have a closed form, so descending to it is a convenience. Two things change that:

- p or n large enough that $X^T X$ cannot be formed or stored

Note: From $\theta^{(0)} = 0$ the iterates never move in a direction that leaves $X\theta$ unchanged, so the limit is the minimizer of smallest norm.

A random handful of rows per step gives stochastic gradient descent.

What gradient descent is really for

Here we have a closed form, so descending to it is a convenience. Two things change that:

- p or n large enough that $X^T X$ cannot be formed or stored
- a model with **no closed form at all**: change $\mathcal{F}$, or change the squared error, and the normal equations have no analogue

Note: From $\theta^{(0)} = 0$ the iterates never move in a direction that leaves $X\theta$ unchanged, so the limit is the minimizer of smallest norm.

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Recall: Fitting one Gaussian

Problem (Gaussian modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that $p(x) = \mathcal{N}(x \mid \mu, \sigma^2)$ **maximizes the likelihood**.

- the log-likelihood is

$$\log L(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

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- it is maximized by

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

New idea: Responses are Gaussians

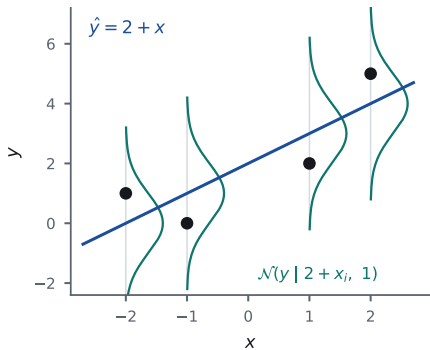
Definition (Gaussian linear model)

Assume the response y_i has the form

$$y_i = \theta^\top x_i + \varepsilon_i,$$

where the ε_i are independent $\mathcal{N}(0, \sigma^2)$ random variables. Equivalently,

$$p(y_i) = \mathcal{N}(y_i \mid \theta^\top x_i, \sigma^2).$$



The **likelihood** of the model

Problem (Gaussian linear model)

Given $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^{p+1}$, find $\theta \in \mathbb{R}^p$ and $\sigma^2 > 0$ such that

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maximizes the log-likelihood $\ell(\theta, \sigma^2)$.

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Surprise: Least squares *is* maximum likelihood

- We found the log-likelihood to be

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Key idea: Minimizing squared error is maximizing the likelihood of a model with Gaussian noise of constant variance.

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Example: More features, better fit?

Least squares on all 442 patients, all features standardized:

	10 features	64 features
training $\text{MSE}(\hat{\theta})$	2860	2417
R^2	0.518	0.592
largest $ \hat{\theta}_j $	37.7	442
$\kappa(X) = \sigma_{\max}/\sigma_{\min}$	21.7	5470

- $\kappa(X)$ is large when some combination of the columns is nearly zero

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Watch out: Every number in this table is computed on the patients used to fit $\hat{\theta}$.

Example: A test set

Fit on 342 random patients, compute the MSE on the other 100, and average over 500 random splits:

	test MSE
constant $\bar{y}$	5946
least squares, 10 features	3010
least squares, 64 features	3805

- 64 features did worse than 10 in 97% of the splits

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	test MSE
constant $\bar{y}$	5946
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- 64 features did worse than 10 in 97% of the splits
- the fit to the training patients improved, and the predictions for new patients got worse

Definition: Ridge regression

Definition (Ridge regression)

For $\lambda > 0$, find $\theta \in \mathbb{R}^p$ minimizing

$$\text{MSE}_\lambda(\theta) + \lambda \|\theta\|^2 = \frac{1}{n} \|y - X\theta\|^2 + \lambda \|\theta\|^2.$$

Convention: columns of X are standardized, y is centered, and there is no column of ones; the intercept is $\bar{y}$.

- Convention puts every coefficient on the same scale, and does not “pull” the intercept toward 0.

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- Convention puts every coefficient on the same scale, and does not “pull” the intercept toward 0.
- Idea is to penalize the large θ_i that arose from having many unhelpful features.

Ridge always has **one** solution

- By gradient rules from last time, $\nabla \text{MSE}_\lambda(\theta) = \frac{2}{n}X^\top(X\theta - y) + 2\lambda\theta$, so
$$\nabla \text{MSE}_\lambda(\theta) = 0 \iff (X^\top X + n\lambda I)\hat{\theta}_\lambda = X^\top y.$$

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- For every $v \neq 0$,

$$v^\top (X^\top X + n\lambda I)v = \|Xv\|^2 + n\lambda \|v\|^2 > 0,$$

so $X^\top X + n\lambda I$ is invertible (**why?**), even when $\text{rank}(X) < p$.

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- As the penalty becomes vanishingly weak, i.e., $\lambda \rightarrow 0$, $\hat{\theta}_\lambda$ approaches the least squares solution $\hat{\theta}$.

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- By gradient rules from last time, $\nabla \text{MSE}_\lambda(\theta) = \frac{2}{n}X^\top(X\theta - y) + 2\lambda\theta$, so

$$\nabla \text{MSE}_\lambda(\theta) = 0 \iff (X^\top X + n\lambda I)\hat{\theta}_\lambda = X^\top y.$$

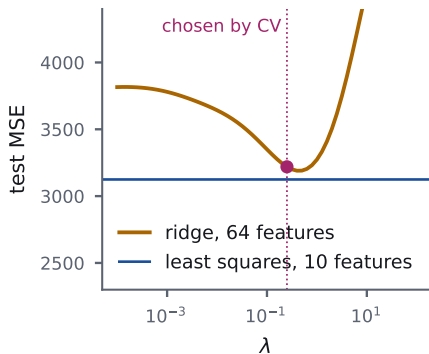
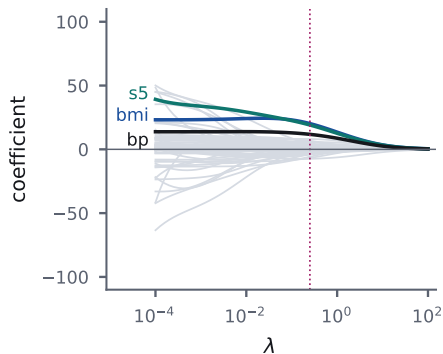
- For every $v \neq 0$,

$$v^\top (X^\top X + n\lambda I)v = \|Xv\|^2 + n\lambda \|v\|^2 > 0,$$

so $X^\top X + n\lambda I$ is invertible (**why?**), even when $\text{rank}(X) < p$.

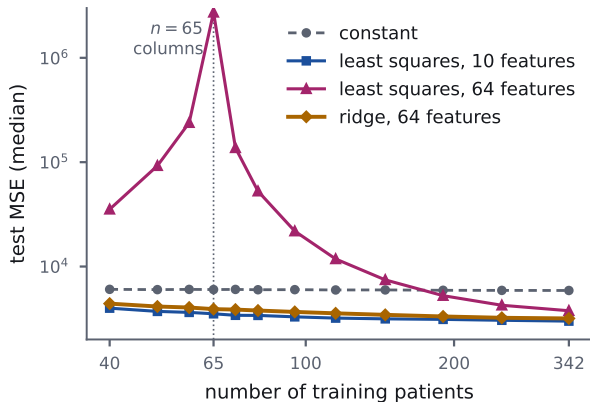
- As the penalty becomes vanishingly weak, i.e., $\lambda \rightarrow 0$, $\hat{\theta}_\lambda$ approaches the least squares solution $\hat{\theta}$.
- As the penalty becomes overwhelmingly strong, i.e., $\lambda \rightarrow \infty$, $\hat{\theta}_\lambda \rightarrow 0$.

Application: Ridge on 64 features



One split, with λ chosen by cross-validation on the 342 training patients: test MSE 3219 (against 3816 as $\lambda \rightarrow 0$), and largest coefficient 20 (against 1932).

Ridge at **every** training size



342 patients, 500 splits

constant	5946
least squares, 10	3010
least squares, 64	3805
ridge, 64	3170

Note: Although ridge regression makes major improvement, the 64 features still do not beat the 10 measurements.